

Table 1. Full HarmBench robust-refusal results for each completed model scale. Direct requests and each of the five frozen human-jailbreak templates contain the same 240 behavior identities.

Model	Template	Method	ASR (%) ↓	Safe-concept relevance (0–2) ↑	Instruction relevance (0–2) ↑	Fluency (0–2) ↑	Overall steering (0–2) ↑
Llama-3.2-1B-Instruct	Direct	Original	10.00	1.91	1.73	1.85	1.62
		A-LQR	9.58	1.95	<u>1.51</u>	<u>1.89</u>	<u>1.46</u>
		H_∞ (ours)	2.92	1.96	1.29	1.74	1.22
	John persona	Original	5.83	1.97	0.68	1.60	0.65
		A-LQR	6.25	1.97	<u>0.62</u>	<u>1.52</u>	<u>0.55</u>
		H_∞ (ours)	1.25	1.98	0.39	1.19	0.34
	DNE nonresponse	Original	6.25	1.99	0.63	1.59	0.64
		A-LQR	4.17	<u>1.99</u>	<u>0.57</u>	<u>1.53</u>	<u>0.56</u>
		H_∞ (ours)	1.67	1.98	0.38	1.18	0.30
	Jailbreak Bot	Original	4.58	1.96	0.64	1.62	0.63
		A-LQR	3.33	1.96	<u>0.56</u>	<u>1.53</u>	<u>0.53</u>
		H_∞ (ours)	0.42	1.98	0.38	1.17	0.30
	YOJA/Nona roleplay	Original	8.33	1.97	0.63	1.56	0.62
		A-LQR	8.75	1.97	<u>0.55</u>	<u>1.50</u>	<u>0.51</u>
		H_∞ (ours)	0.83	1.98	0.37	1.15	0.29
	APM programmer	Original	2.92	1.94	0.64	1.56	0.64
		A-LQR	5.00	1.97	<u>0.57</u>	<u>1.51</u>	<u>0.53</u>
		H_∞ (ours)	0.42	1.98	0.37	1.12	0.30
Llama-3.2-3B-Instruct	Direct	Original	16.67	1.88	1.21	1.88	1.19
		A-LQR	20.00	1.91	1.18	1.86	1.15
		H_∞ (ours)	8.33	1.99	1.19	1.90	1.18
	John persona	Original	11.67	1.90	0.61	1.54	0.58
		A-LQR	11.67	<u>1.96</u>	<u>0.69</u>	1.52	<u>0.63</u>
		H_∞ (ours)	4.17	1.94	0.60	1.65	0.59
	DNE nonresponse	Original	10.83	1.93	0.57	1.52	0.55
		A-LQR	10.42	<u>1.95</u>	<u>0.63</u>	1.52	<u>0.59</u>
		H_∞ (ours)	3.75	1.95	0.59	1.66	0.58
	Jailbreak Bot	Original	14.17	1.90	0.57	1.54	0.56
		A-LQR	12.92	<u>1.97</u>	<u>0.64</u>	1.51	<u>0.60</u>
		H_∞ (ours)	3.75	1.93	0.59	1.67	0.58
	YOJA/Nona roleplay	Original	11.25	1.91	0.59	1.52	0.55
		A-LQR	11.67	<u>1.95</u>	<u>0.65</u>	1.51	<u>0.59</u>
		H_∞ (ours)	4.17	1.94	0.56	1.64	0.56
	APM programmer	Original	10.83	1.91	0.56	1.48	0.53
		A-LQR	11.25	<u>1.97</u>	<u>0.65</u>	1.47	<u>0.60</u>
		H_∞ (ours)	2.50	1.96	0.58	1.60	0.58